Setting Up and Using a Telescope: A Practical Guide

What you will be able to do

1. Assemble and balance a telescope safely.
2. **Align the finder** with the main telescope.
3. Use an alt-az / Dobsonian mount and roughly **polar-align an equatorial mount** from India.
4. Focus, choose eyepieces and find your first targets.
5. Check **collimation** of a Newtonian and solve common problems.
6. Care for your equipment in Indian conditions — dust, humidity, heat.



A 200 mm Newtonian reflector on a German equatorial mount being set up at a star party. Note the counterweight and the polar axis tilted towards the pole.

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1. Before the night: choosing your spot

- **Level, firm ground**, away from walls and roofs that radiate heat (concrete rooftops stay warm long after sunset and make images shimmer).
- A clear view of the part of the sky you want — plus north if you need to polar-align.
- Shielded from street lights; even a building's shadow helps.
- Bring the telescope out **30–60 minutes early** so the optics cool to the night air (most important for reflectors).

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P.T.O.

2. Assembly checklist (do it once in daylight first!)

1. **Tripod:** extend and lock the legs, spread them fully, and level the head (many mounts have a bubble level).
2. **Mount head:** attach it. On an equatorial mount, fit the **counterweight shaft and counterweights FIRST**, before the tube, so it cannot tip over.
3. **Tube:** attach it with the clamp or dovetail; keep one hand on it until it is secure.
4. **Finder**, then **diagonal and eyepiece** — start with the **lowest power** (longest focal length, e.g. 25 mm).
5. **Balance** (equatorial): unlock the RA axis and slide the counterweight until the tube stays put in any position; then unlock Dec and slide the tube in its rings until it balances front to back.

3. Align the finder — the step beginners skip

In daytime, on a distant object (a mobile tower, temple spire or far tree) — NEVER the Sun:

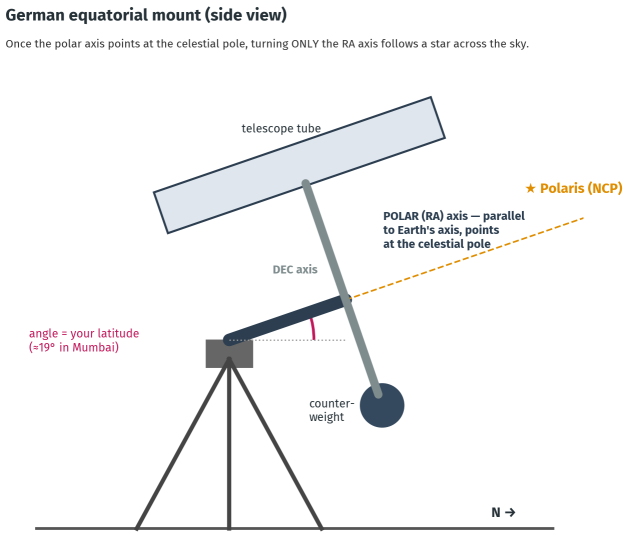
1. Centre the object in the main telescope with the low-power eyepiece.
2. Adjust the finder's screws until its crosshair (or red dot) is on the **same** object.
3. Switch to higher power, re-centre and fine-tune. At night, check on a bright star or the Moon.

Now whatever is in the finder's crosshair will be in your eyepiece.

4. Alt-azimuth and Dobsonian telescopes

Simply **point and look**, nudging the tube to follow. With a 52° eyepiece at 100× the true field is about 0.5°, and an object near the celestial equator drifts across it in about **2 minutes** because the sky turns 15″ every second. Tip: place the object at the edge of the field it is about to enter and let the sky carry it across.

5. Equatorial mount: rough polar alignment from India



The polar axis must be parallel to Earth's axis, pointing at the north celestial pole near Polaris.

1. **Set the latitude scale** to your latitude (19° Mumbai, 28.6° Delhi, 13° Bengaluru). Some cheap mounts struggle below ~ 15°; tilt the tripod slightly if needed.

2. **Point the polar axis due north** — at Polaris if visible, or with a phone compass (India's magnetic declination is only $\sim 0°$–$2°$).
3. **Fine-tune** with the altitude and azimuth bolts until Polaris is centred. With a polar scope, an app such as "Polar Scope Align" shows where Polaris should sit in its reticle.
4. For **visual use, "roughly north" is enough**: turn only the RA knob to follow objects. For astrophotography use drift alignment or camera software.

South India

Polaris is only 8°–13° high and often hidden by haze or buildings. Use the compass and latitude scale, then refine by drift alignment. (The southern pole star, σ Octantis, is never visible from India.)

Moving an EQ mount: RA follows stars east–west; Dec moves north–south. Rotate the *tube in its rings* (not the mount) to bring the eyepiece to a comfortable angle. **GoTo mounts:** after polar alignment, do the hand controller's 2–3-star alignment and enter the correct **date, time (IST = UTC + 5:30) and location** — wrong time-zone settings are the most common GoTo error.

6. Focusing and choosing magnification

1. Start with low power; point at the Moon or a bright star; turn the focuser slowly until a star is the **smallest possible point**.
2. Centre first, *then* change to higher power; refocus each time.
3. Glasses: observe without them for short/long sight, but keep them on for astigmatism.
4. Use **averted vision** for faint objects, and gently tap the tube — movement helps the eye notice faint things.

7. Your first night: easy targets from India

Target	When	Eyepiece	What you'll see
The Moon	Shukla Chaturthi to Ekadashi	25 mm, then 10 mm	craters and mountains along the terminator
Jupiter	evenings Feb–Jul 2027 (opposition Feb, in Leo)	10 mm	cloud belts; 4 moons that change nightly
Saturn	evenings Oct 2026–Mar 2027	10 mm (+ Barlow)	**the rings!** and the moon Titan
Mars	evenings Feb–Jun 2027 (opposition Feb)	10 mm + Barlow	small orange disc; polar cap on good nights
Venus	mornings Nov 2026–May 2027	25 mm	phases like a small Moon
Pleiades	Oct–Mar	25 mm / binoculars	dozens of blue stars
Orion Nebula	Nov–Mar	25 mm	glowing gas with 4 young "Trapezium" stars
Mizar & Alcor	Mar–Jun	25 mm	a double — and Mizar itself splits in two
Albireo	Jul–Nov	25 mm	a gold and blue pair
Hercules cluster	May–Aug	10 mm	a ball of thousands of stars

(continued)

Target	When	Eyepiece	What you'll see
Omega Centauri	Apr–Jun, low south	25 mm	the biggest globular cluster — best from south India

8. Collimation: is your Newtonian aligned?

Mirrors can shift slightly, especially after bumpy journeys, giving fuzzy images at high power. **Star test:** on a steady night centre a bright star at high power and *defocus slightly*. You should see **concentric rings** with the shadow of the secondary mirror in the centre. A lopsided "doughnut" means collimation is needed. **Basic collimation** (with a collimation cap or laser): first adjust the secondary mirror so it is centred under the focuser and shows the whole primary; then adjust the primary's three screws until the reflection of the secondary is centred. Ask your instructor to demonstrate first.

9. Troubleshooting

Problem	Likely cause	Fix
Nothing / only black	lens cap on; object not in view	remove cap; low power; check finder alignment
Blurry grey blob	out of focus	turn the focuser fully one way, then slowly back
Image upside down / reversed	normal for astronomical telescopes	doesn't matter in space!
Dark "doughnut" when defocused	normal for reflectors	focus properly
Image boils and shimmers	poor seeing, warm tube or rooftop	let it cool; observe from grass; lower power
Optics fog up	dew (coastal India, winter)	dew shield; never wipe a wet lens; let it dry indoors with caps off
Object races out of view	too much magnification	lower power; on EQ turn only the RA axis
GoTo points wrongly	wrong date/time/time-zone/location	re-enter IST (UTC + 5:30), check AM/PM, re-align

10. Caring for your telescope in Indian conditions

- **Dust:** keep caps on; store under a cover; clean optics rarely (blower first, proper lens fluid only when really needed).
- **Humidity and fungus** (monsoon!): store with **silica-gel packets**; never store a telescope wet.
- **Heat:** never leave it in a hot car or direct sun. **Transport:** pad the tube; re-check collimation after long trips.

⚠ The golden safety rules

1. **NEVER point a telescope, finder or binoculars at the Sun** without a certified solar filter covering the **front** (and the finder capped).
2. Supervise children near a telescope in daytime.

⚠ The golden safety rules *(continued)*

3. Keep fingers clear of moving GoTo mounts and counterweights.

Workshop checklist (signed by the instructor)

Skill	Done ✔	Instructor's initials
Assembled and levelled the tripod and mount		
Balanced an EQ mount (both axes)		
Aligned the finder in daylight		
Set the latitude and pointed the polar axis north		
Found and focused the Moon or a planet		
Calculated the magnification of each eyepiece used (Lecture 3.3)		
Located one deep-sky object		
Demonstrated the star test		

📖 Further reading (free / public)

- G. Consolmagno & D. M. Davis, *Turn Left at Orion* (Cambridge University Press) — what you will really see in a small telescope.
- T. Dickinson & A. Dyer, *The Backyard Astronomer's Guide* (Firefly Books) — choosing and using equipment.